Multi-Objective Optimization

Lecture 8 - Management Science

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Introduction

Client Briefing: EcoExpress

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Operations Director's Dilemma:

"EU regulations demand 40% emission cuts, but we can't sacrifice profitability, service quality, or reliability!"

The Fleet Challenge

EcoExpress operates regional last-mile delivery across 3 cities

- EU Green Deal: 40% emission reduction by 2025
- Rising fuel costs (€2.1/L diesel)
- Amazon entering our market (speed pressure)
- Driver shortage (need automation-friendly vehicles)

. . .

Question: How do we transform our fleet while staying competitive?

Today's Learning Objectives

By the end of this lecture, you will be able to:

1. Explain why most decisions involve competing objectives
2. Identify and visualize Pareto optimal solutions
3. Apply normalization techniques to make objectives comparable
4. Implement apporaches to find trade-off solutions
5. Make decisions from a Pareto frontier

Quick Recap: Local Search

Last week we optimized routes for delivery:

- Started with greedy construction (e.g. Nearest Neighbor)
- Improved with local search (e.g. 2-opt)
- Considered time windows
- **But: We only optimized distance**

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Question: What if we also care about emissions, cost, AND customer satisfaction?

The Problem

Single vs Multi-Objective

Single Objective

- “Minimize total distance”
- Clear winner. Easy, right!

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Multiple Objectives

- “Minimize cost AND emissions AND maximize speed”
- No clear answer…

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Question: Any idea how to approach this?

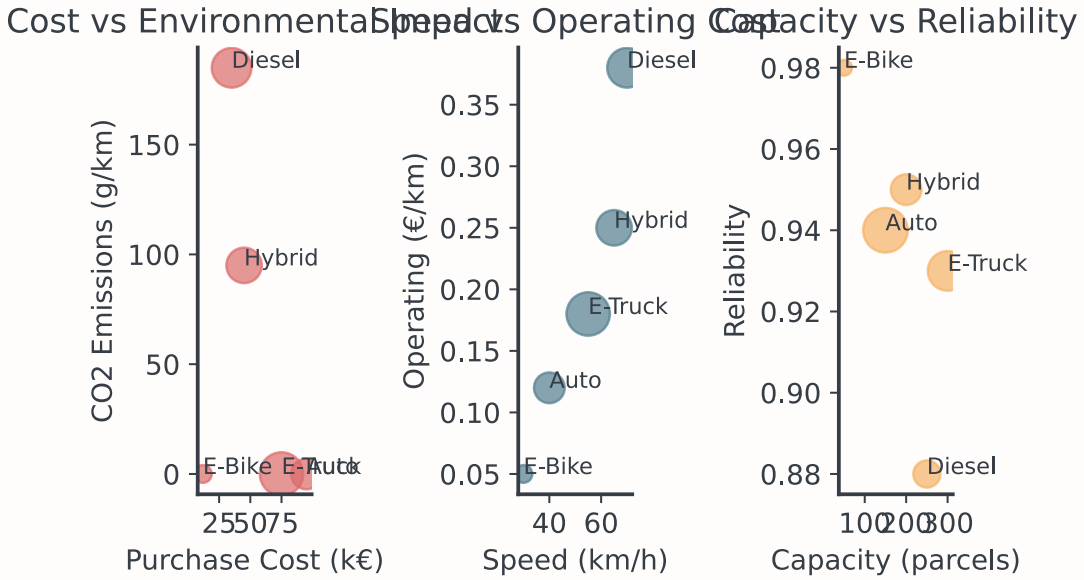
EcoExpress Vehicle Options

Type	Purchase Cost (€)	Operating (€/km)	CO2 (g/km)	Speed (km/h)	Capacity (parcels)	Reliability
E-Truck	75000	0.18	0	55	300	0.93
Hybrid	45000	0.25	95	65	200	0.95
Diesel	35000	0.38	185	70	250	0.88
E-Bike	12000	0.05	0	30	50	0.98
Auto	95000	0.12	0	40	150	0.94

. . .

Question: Which vehicle is “best” for EcoExpress?

Trade-offs Everywhere



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> **!** Important
>
> Every vehicle excels at something different!

Real Business Constraints

Beyond the numbers, consider:

- **EU regulations:** Carbon tax of €100/ton CO_2 starting 2025
- **Competition:** Amazon promises 2-hour delivery
- **Labor market:** Autonomous vehicles reduce driver dependency
- **Urban zones:** Zero-emission zones in city centers
- **Peak times:** Black Friday = 3x normal volume

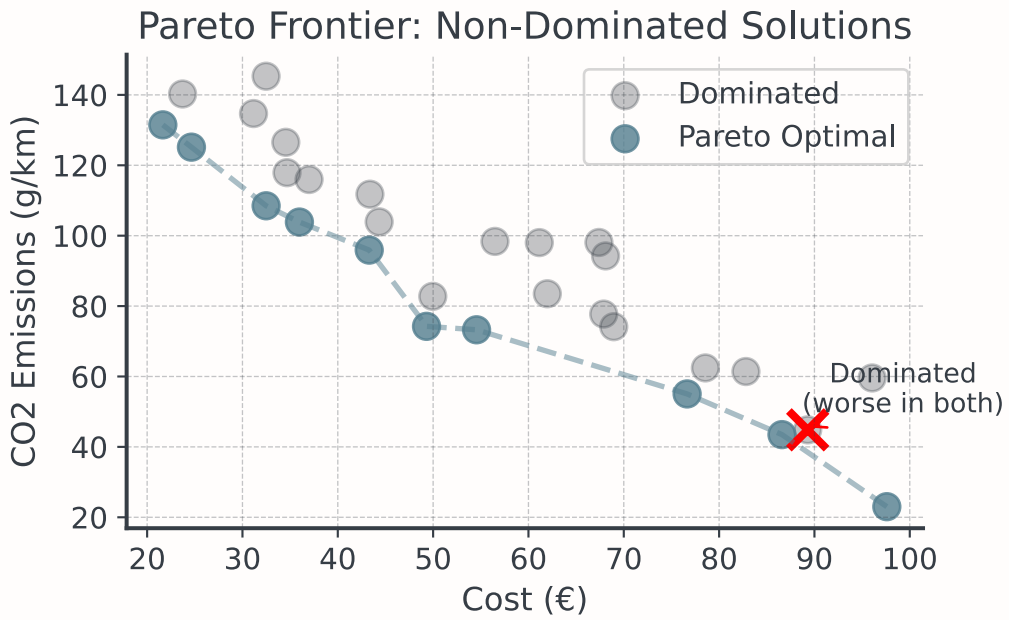
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> **!** Important
>
> There is no single “optimal” solution - only trade-offs

Pareto Optimality

Dominated Solutions

A solution is dominated if another solution is:



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> ! Important
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> Better in at least one objective and not worse in any objective!

The Pareto Frontier

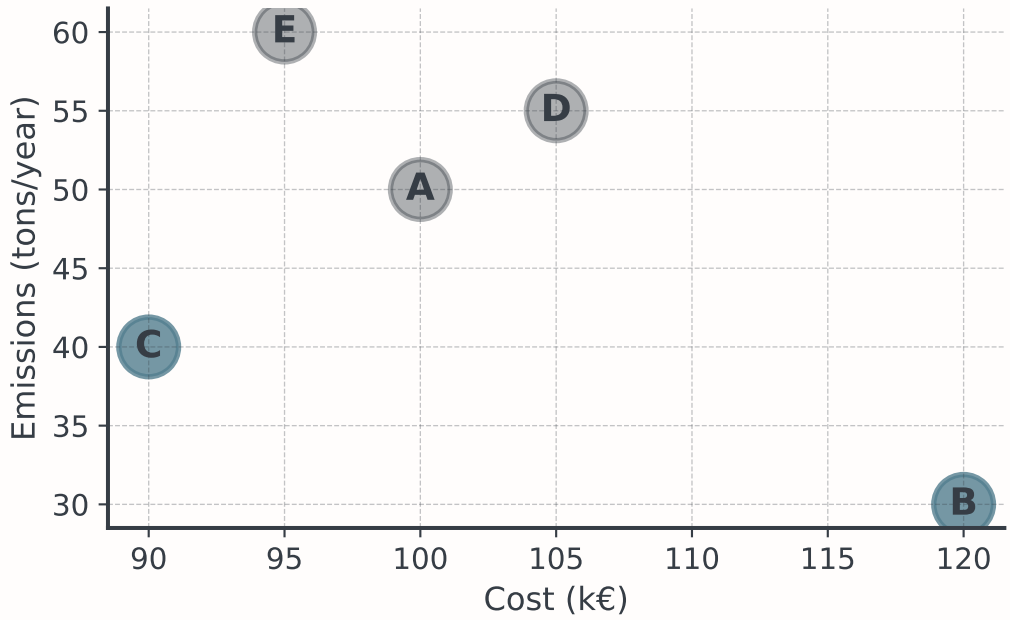
The Pareto frontier is the set of all non-dominated solutions

- No solution is objectively "better"
- Each represents a different trade-off
- Moving along frontier: gain in one objective, loss in another
- Decision makers choose based on **preferences**

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Question Do you think you get the idea?

Find the Non-Dominated

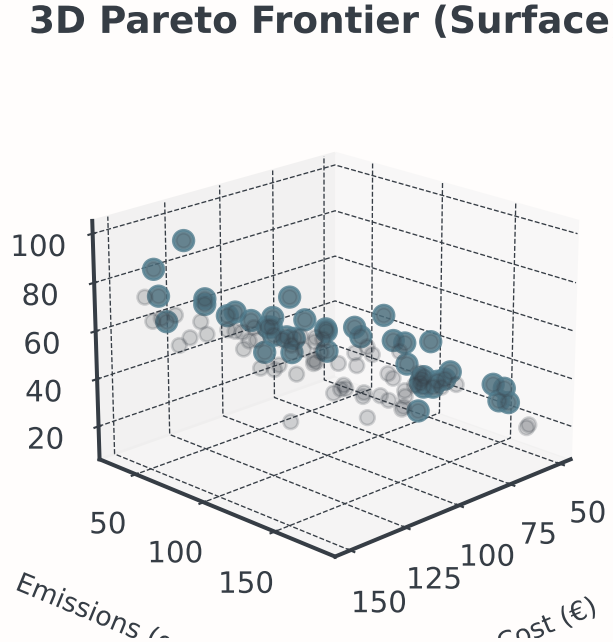


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Question: Which fleets are non-dominated?

Three+ Objectives

With 3 objectives, the Pareto frontier becomes a surface:



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! Important

Harder to visualize, but same principle applies!

Fleet Composition

The Fleet Challenge

EcoExpress needs to replace their 80 diesel vans

- Must meet **EU regulation**: Average emissions ≤ 111 g CO_2/km
- Need capacity for **22,000 parcels/day**
- Must balance **cost vs. service quality**
- 5 vehicle types available, each with trade-offs

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Question: How do we choose the right mix?

Vehicle Options Recap

Type	Purchase Cost (€)	Operating (€/km)	CO2 (g/km)	Speed (km/h)	Capacity (parcels)	Reliability
E-Truck	75000	0.18	0	55	300	0.93
Hybrid	45000	0.25	95	65	200	0.95
Diesel	35000	0.38	185	70	250	0.88
E-Bike	12000	0.05	0	30	50	0.98
Auto	95000	0.12	0	40	150	0.94

. . .

Notice: No single vehicle is “best” at everything!

Fleet Composition Framework

This is a discrete selection problem, not continuous allocation

Decision Variables:

- Fleet: How many of each vehicle type? (discrete/integer)
- n_i = number of vehicles of type i (integers!)
- Example: $n_{\text{E-Truck}} = 20$, $n_{\text{Hybrid}} = 30$, etc.

Objective 1: Total Cost

Purchase cost + Operating cost over 3 years

$$\text{Total Cost} = \sum_i n_i \cdot (P_i + O_i \cdot d \cdot y)$$

- n_i = quantity of vehicle type i
- P_i = purchase cost of vehicle type i
- O_i = operating cost per km for type i
- d = daily distance × days per year
- y = years

Objective 2: Service Score

Composite measure of fleet performance

$$\text{Service Score} = 0.5 \cdot C_{\text{score}} + 0.3 \cdot R_{\text{score}} + 0.2 \cdot S_{\text{score}}$$

- $C_{\text{score}} = \min\left(1.0, \frac{\text{Total Capacity}}{22000}\right)$ (capacity adequacy)
- $R_{\text{score}} = \frac{\sum n_i \cdot r_i}{\sum n_i}$ (weighted avg. reliability)
- $S_{\text{score}} = \frac{\sum n_i \cdot s_i}{70 \cdot \sum n_i}$ (normalized speed)

...

> **i** Note
>
> Service score captures multiple performance dimensions in one metric!

Hard Constraint: Emissions

EU regulation creates a feasibility boundary

$$\text{Average } CO_2 = \frac{\sum_i n_i \cdot e_i}{\sum_i n_i} \leq 111 \quad \text{g/km}$$

Where e_i = CO_2 emissions per km for vehicle type i

...

This eliminates some solutions:

- All diesel vans: 185 g/km > 111
- Mix with too many diesel: Still violates
- Zero-emission + some diesel: Might work

Data Source

Where Do These Numbers Come From?

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Vehicle Specifications:

- **Purchase costs**: Manufacturer quotes, market research
- **Operating costs**: Fuel/electricity prices, maintenance records
- **Capacity**: Vehicle specs (cargo volume, weight limits)

- **Reliability**: Historical uptime data, manufacturer warranties
- **EU Standards**: WLTP certification for vehicles
- **Electric vehicles**: Grid carbon intensity (kWh → g CO_2)

Example Fleet Comparison

```
Three Fleet Strategies:

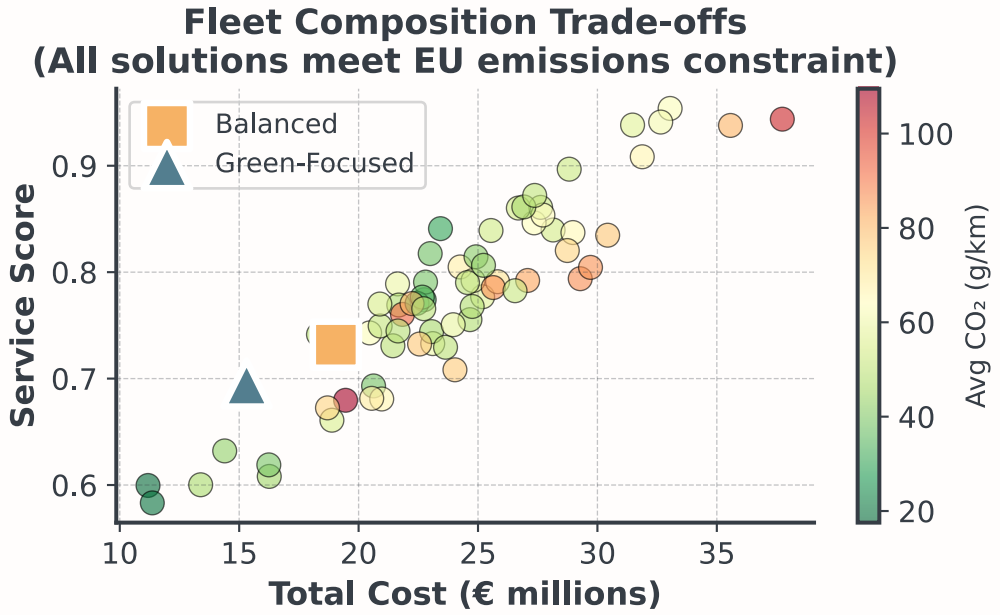
         name    cost  service        co2  capacity  vehicles
 Cost-Focused 28.9996 0.809705 120.714286     15000        70
     Balanced 19.0478 0.731840  33.928571     13250        70
Green-Focused 15.3102 0.695373   0.000000     12750        75


Cost-Focused: ✗ VIOLATES (CO2: 120.7 g/km)
Balanced: ✓ Compliant (CO2: 33.9 g/km)
Green-Focused: ✓ Compliant (CO2: 0.0 g/km)
```

...

Question: Which strategy would you choose?

Visualizing Fleet Trade-offs




```
Generated 68 feasible fleet compositions
```

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> ! Important
>
> Each point is a different fleet mix, all meeting emissions constraint!

Solution Approaches

Multi-Objective Optimization

You can use optimization solvers or heuristics!

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With Optimization Solvers

- Weighted Sum Method
- ε-Constraint Method
- Goal Programming
- **Optimal solutions**
- **Need mathematical model**

With Heuristics

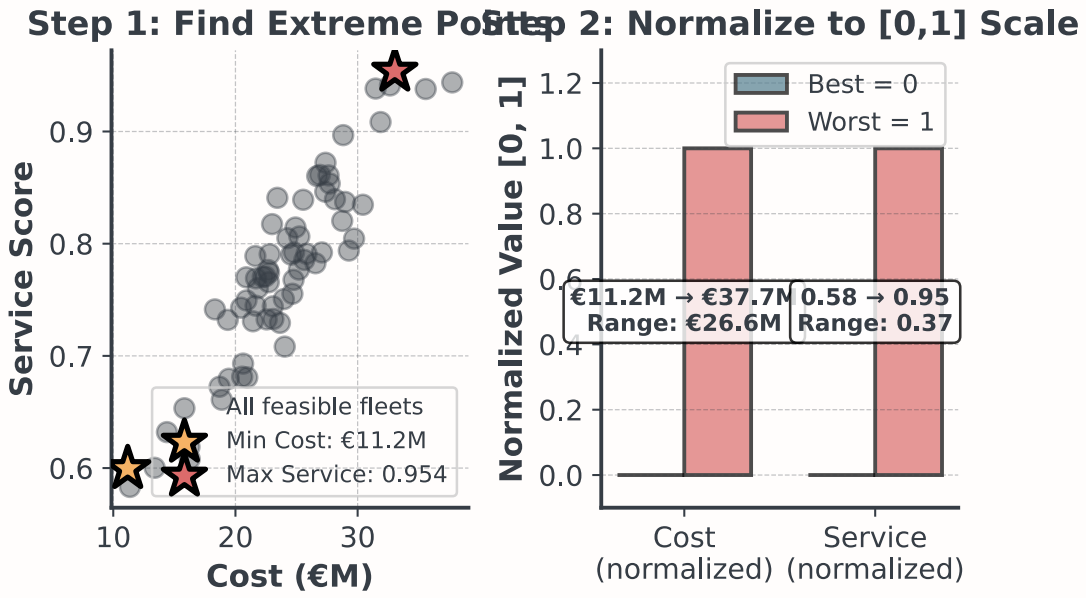
- Weighted Greedy Construction
- Multi-Objective Local Search
- Metaheuristics
- **Good solutions, fast**
- **No optimality proof**

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> ! Important
>
> In this lecture we use heuristic approaches!

Foundation: Extreme Points

First step for BOTH approaches - find the boundaries:

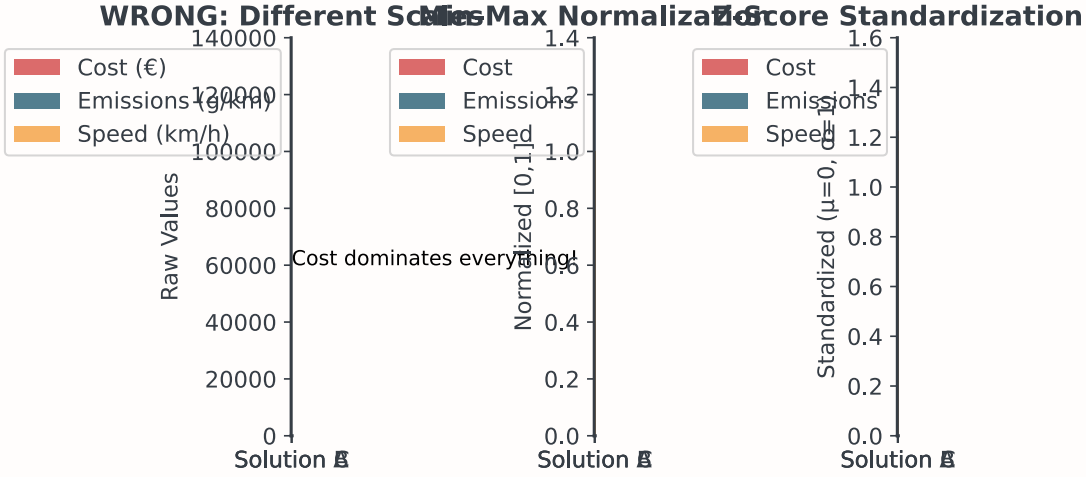


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Question: Why is normalization essential?

Critical: Normalization

Without it, your analysis is meaningless



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Question: Any intuition on how to do [0,1] normalization?

How to Normalize

The Normalization Formula for [0,1]

$$\text{Normalized}_i = \frac{x_i - x_{min}}{x_{max} - x_{min}}$$

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In Python, this is rather simple!

...

```
def normalize_objectives(data):
    return (data - data.min()) / (data.max() - data.min())

# Now weights actually mean something
weighted_score = w1 * normalize(cost) + w2 * normalize(emissions)
```

...

> Tip
>
> Easy, right?

Extreme Points

There are several reasons why extreme points matter:

1. **Trade-off Space**: Min/max values bound your Pareto frontier
2. **Enable Proper Normalization**: Need ranges for scaling to [0,1]
3. **Feasibility**: If single objectives not achievable, problem infeasible
4. **Stakeholder**: “Best cost is €50k, best emissions is 40kg”

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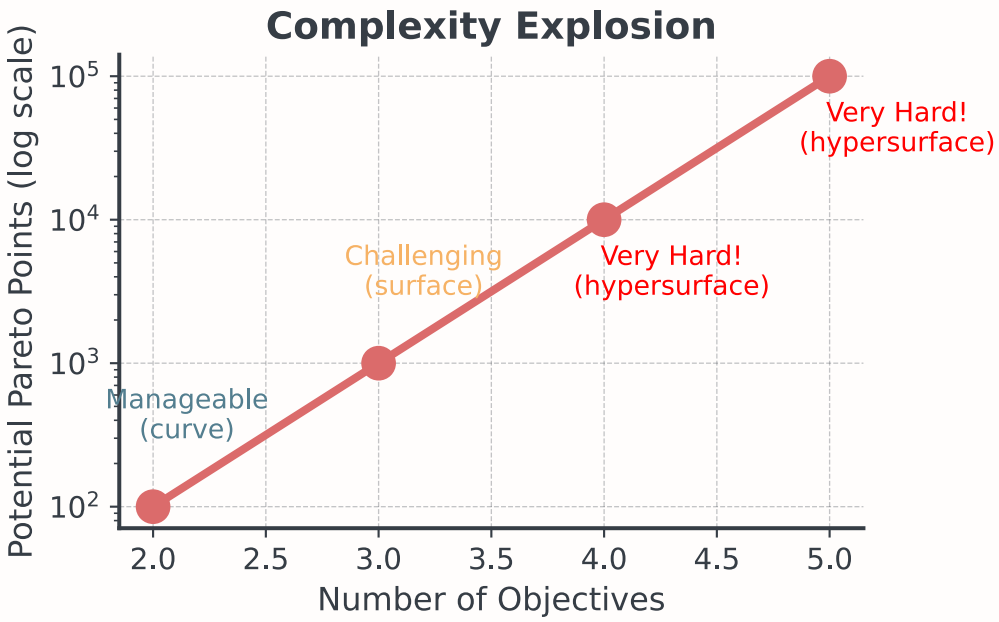
Implementation Pattern:

```
def find_extreme_points(problem):
    # Solve for minimum cost (ignore emissions)
    min_cost_solution = minimize(cost_objective, constraints)
    # Solve for minimum emissions (ignore cost)
    min_emissions_solution = minimize(emissions_objective, constraints)
```

Computational Complexity

How hard does it get with more objectives?

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> Tip
>
> Why? Because there are just way more potential solutions to check!

Solver-Based Methods

Quick overview - you won't implement these in assignments

1. **Weighted Sum:** Minimize $w_1 \times \text{cost} + w_2 \times \text{emissions}$
   - Simple, fast for convex problems
2. **ε-Constraint:** Minimize cost subject to emissions $\leq \varepsilon$
   - Systematically vary ε to find complete frontier
3. **Goal Programming:** Minimize deviations from targets
   - Set target for each objective, minimize weighted deviations

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> Note
>
> **For your fleet optimization:** You'll use **heuristic approaches** instead!

Heuristic Approach

The Heuristic Strategy

For problems without mathematical models

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1. **Construction:** Build initial solutions with weighted greedy
2. **Improvement:** Multi-objective local search
3. **Selection:** Filter dominated solutions to find Pareto frontier

. . .

> ! Important
>
> **Key difference from solvers:**
>
> - **Solvers:** Need mathematical model, guarantee optimality
> - **Heuristics:** Work with any evaluation function, find good solutions fast

Why Heuristics?

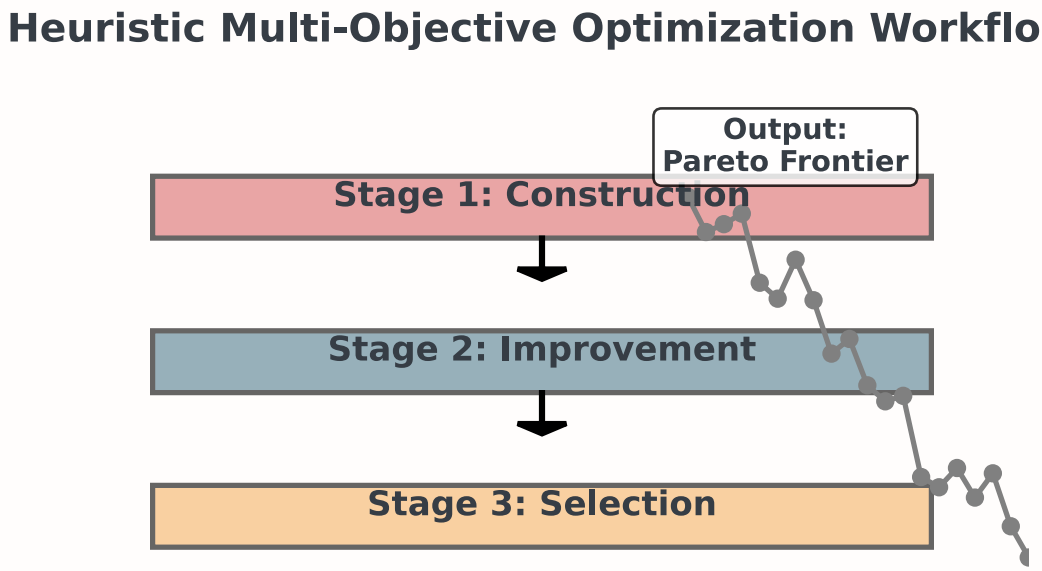
Depending on the problem:

- **Combinatorial explosion**
- Huge solution space even for one problem
- Evaluating one solution might thus take too long
- Need diverse Pareto frontier, not just one “optimal” solution
- Open Source Solvers too slow
- Commercial solvers too expensive

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Question: How do we build good solutions without a solver?

The Three-Stage Heuristic Process



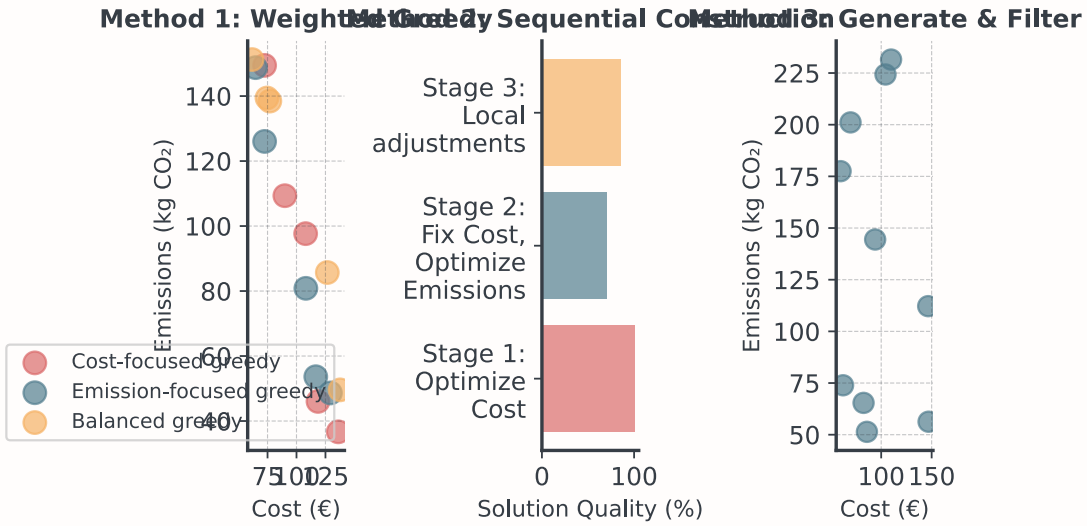
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This is what you'll implement in your assignments!

Construct & Improve

Construction Methods for MOO

How to build initial solutions when you have multiple objectives?



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> **Note**
>
> Three choices (for starters). Let's check them out!

Weighted Greedy Construction

Making greedy choices on a weighted objective

1. **Choose weight vector** $w = (w_1, w_2)$
2. **At each step, pick the choice that minimizes:**

$$w_1 \cdot \text{cost}(x) + w_2 \cdot \text{emissions}(x)$$

3. **Build complete solution greedily**
4. **Repeat** with different weights to explore frontier

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> **Tip**
>
> Different weights explore different trade-offs! Easy, right?

Sequential Greedy (Lexicographic)

Optimize one objective at a time, in priority order

1. **Rank objectives by priority**
   - E.g. cost (most important) and then emissions (tie-breaker)
2. **At each step:**
   - Find choices that minimize **primary** objective
   - If tie → use **secondary** objective
3. **Build one working solution**

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> Tip
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> We could also accept primary values within 10% of best so secondary has more influence!

Diverse Starting Pool

Generate many random solutions, keep the non-dominated ones

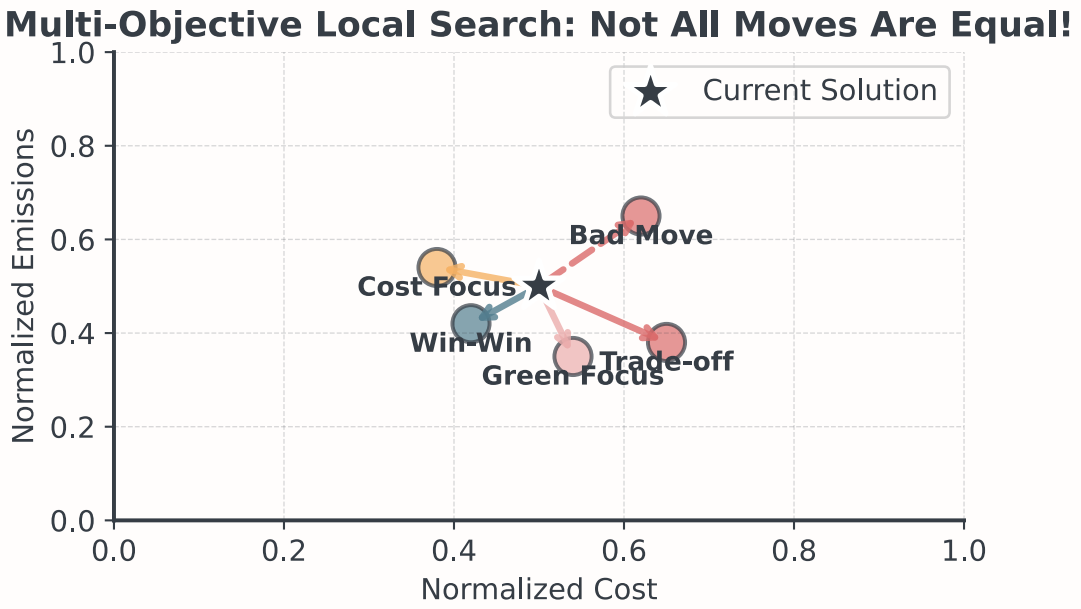
1. **Generate N random solutions (e.g., N=100)**
2. **Evaluate all solutions on both objectives**
3. **Filter to keep only non-dominated solutions**
4. **Result:** A diverse set of Pareto-optimal solutions

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> Tip
>
> - Explores **entire** solution space
> - No bias toward specific weights
> - Great for **warm-starting** local search

Local Search for Multi-Objective

Special moves that improve multiple objectives:



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Question: Which moves are acceptable?

MOO Local Search Rules

Accept a move if:

1. **Dominance**: New solution dominates current (win-win!)
2. **Trade-off**: Improves primary, acceptable loss in secondary
3. **Probabilistic**: Use temperature (like simulated annealing)

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> ! Important
>
> Always keep all your objectives in mind when making decisions.

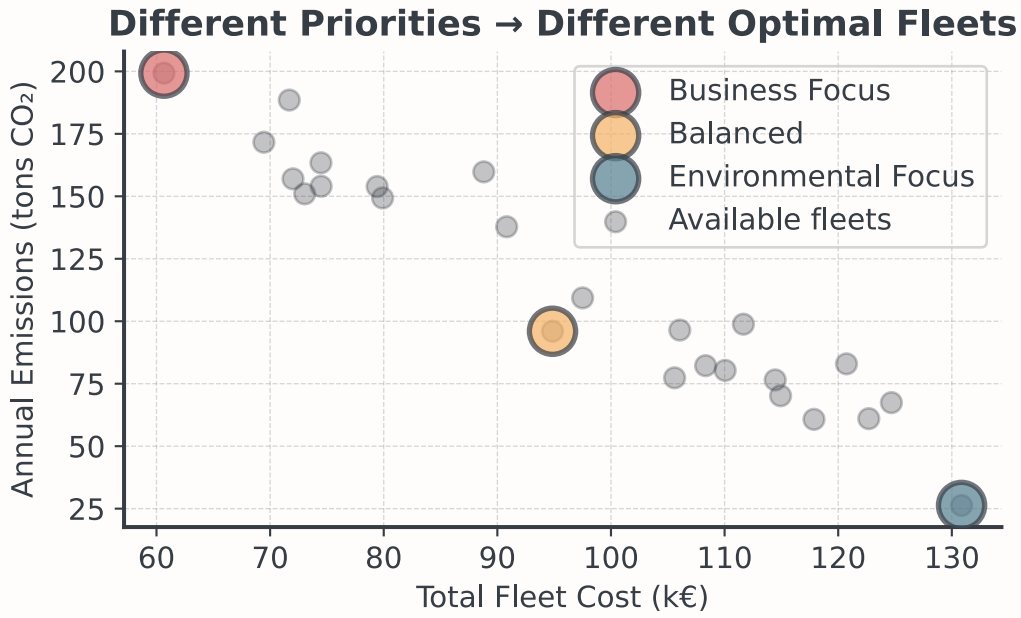
Pareto to Decision

How to Choose!

1. The Knee Point: Find the “elbow” where improvement slows
2. Satisficing Levels: Set minimum acceptable thresholds
   - **Cost** must be < €100k (budget constraint)
   - **Emissions** must be < 100 kg (regulatory limit)
   - **Service level** must be > 90% (customer requirement)
3. Stakeholder Preferences: Let business priorities guide
   - **Sustainability:** Minimum emissions that meets constraints
   - **Operations:** Maximum service level within budget

Weighting has an Impact

The weights thus reflect your values!



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> Tip
>
> Depending on your weight, the choice will vary.

Advanced

Speed vs Sustainability Dilemma

The Three-Way Trade-off in E-Commerce

1. **Minimize Delivery Time** (1-day/2-hour promise)
2. **Minimize Cost** (fuel, labor, fulfillment)
3. **Minimize Environmental Impact** (carbon footprint)

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Faster delivery = More vehicles less full = Higher emissions

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Question: What could retailers do?

Moving the Frontier

Instead of point on the frontier, move the entire frontier:

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Question: Any idea of examples?

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> Tip
>
> R&D can fundamentally change what's possible!

Briefing

Today

Hour 2: This Lecture

- Multi-objective
- Pareto optimality
- Weighted greedy
- Local search MOO

Hour 3: Notebook

- Bean Counter CEO
- Find Pareto frontier
- Apply weighted greedy
- Normalize objectives

Hour 4: Competition

- Fleet composition
- Vehicle selection
- Cost vs service
- Justify choice!

The Competition Challenge

EcoExpress Sustainable Fleet Design

. . .

1. **Select** optimal fleet mix (5 vehicle types)
2. **Balance** cost vs. service score
3. **Meet** EU emission constraint (≤ 111 g CO_2/km)
4. **Ensure** sufficient capacity (22,000 parcels/day)

. . .

> ! Important
>
> Find the best trade-off for your business priorities!

Choosing Your MOO Approach

Different situations call for different methods:

Situation	Best	Why
Clear priorities	Sequential greedy	Fast, hierarchy
Exploring	Weighted greedy	Different solutions
Many solutions	Diverse pool	Builds frontier
Quick solution	Single weighted	One good compromise
Improve existing	Multi-objective local	Refines trade-offs

. . .

Tip

Competition? Generate diverse pool or weigted, then improve with local search.

Implementation Pitfalls to Avoid

Common bugs that cost you time:

1. **Forgetting to Normalize**
   - Always normalize to [0,1] first!
2. **Optimizing Too Many Objectives**
   - 2-3: Manageable, 4+: Exponentially harder
   - Combine related objectives or use constraints
3. **Not Checking Solution Feasibility**
   - Always verify constraints after optimization

Summary

Key Takeaways:

- Real decisions have multiple conflicting objectives
- Pareto frontier shows all rational trade-offs
- Normalization is essential for fair comparison
- Weights reflect values, make them explicit
- Visualization crucial for decision-making

Break!

Take 20 minutes, then we start the practice notebook

Next up: You'll become Bean Counter's expert

Then: The Sustainability competition

Bibliography